TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Macroarea di Ingegneria
Dipartimento di Ingegneria Civile e Ingegneria Informatica

Introduction to Distributed Systems

Sistemi Distribuiti e Cloud Computing
A.A. 2025/26

Valeria Cardellini

Laurea Magistrale in Ingegneria Informatica

Advances in technology and software

Networking

Computing power

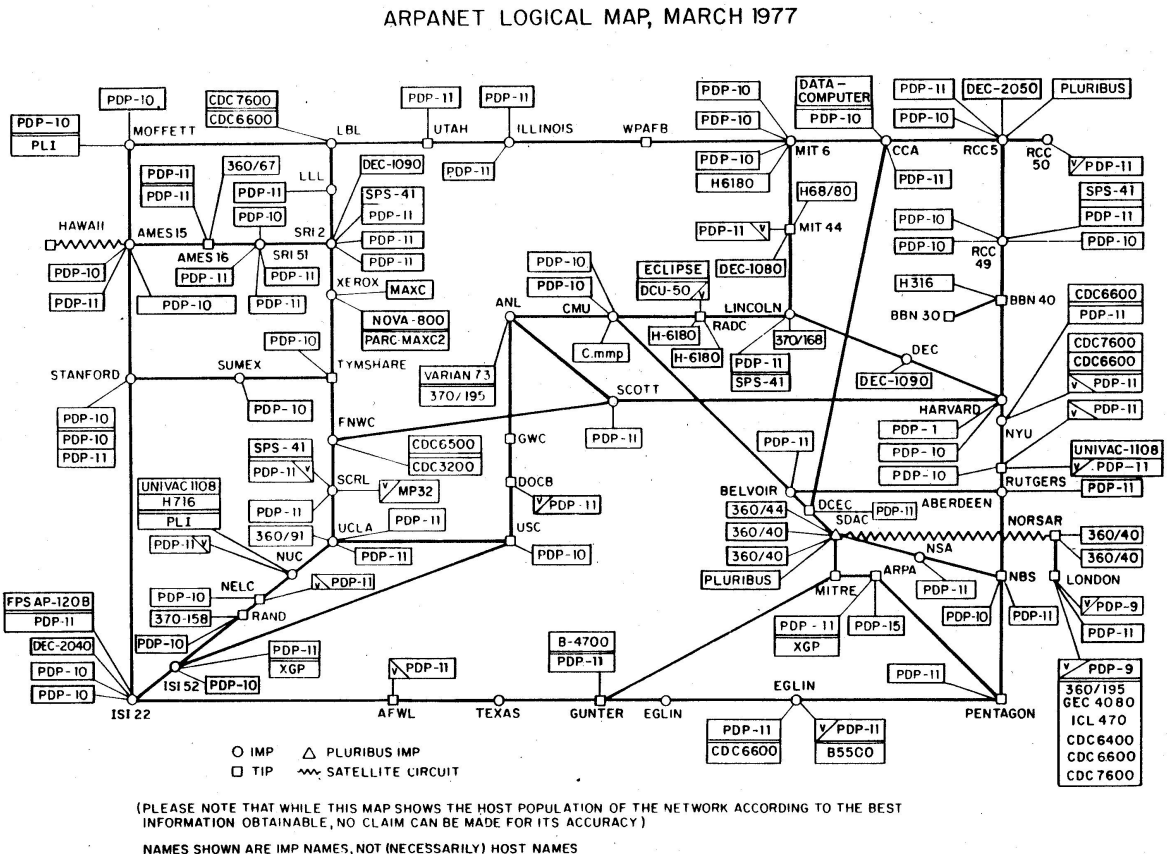
Storage

Memory

Protocols

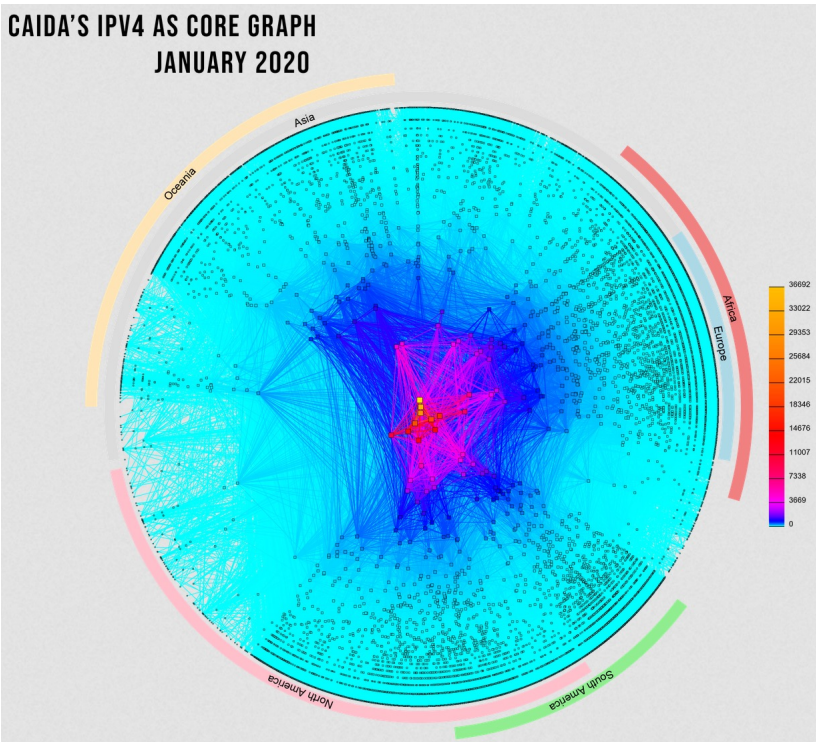
Software services and applications

Internet evolution: 1977



Internet evolution: after 43 years (2020)

- IPv4 AS-level Internet graph
- Interconnections of ~47000 ASs, ~150K links



Source: https://www.caida.org/projects/as-core/

Internet traffic in 2024

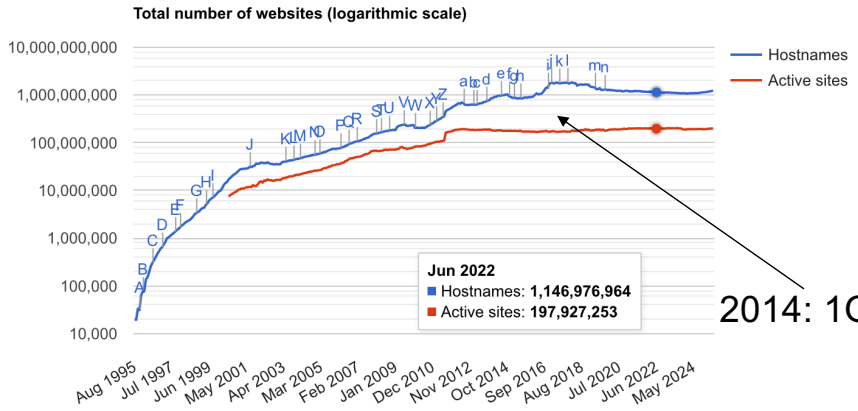
	BRAND	Fixed	Mobile
1	Alphabet	18%	20%
2	Meta	11%	22%
3	Netflix	9%	5%
4	Microsoft	7%	2%
5	Tik Tok	6%	12%
6	Apple	6%	5%
7	Amazon	5%	1%
8	Disney	4%	2%
		66%	**69%**

App Category and App	% of Users	User Volume
Social Media		
Facebook	**35%**	**190 MB**
Tik Tok	**36%**	**173 MB**
Instagram	**35%**	**961 MB**
Snapchat	**25%**	**119 MB**
X (Twitter)	**16%**	**71 MB**
Communication		
WhatsApp	**34%**	**36 MB**
Facebook Messenger	**38%**	**28 MB**
Telegram	**7%**	**74 MB**
Discord	**2%**	**373 MB**
FaceTime	**8%**	**26 MB**

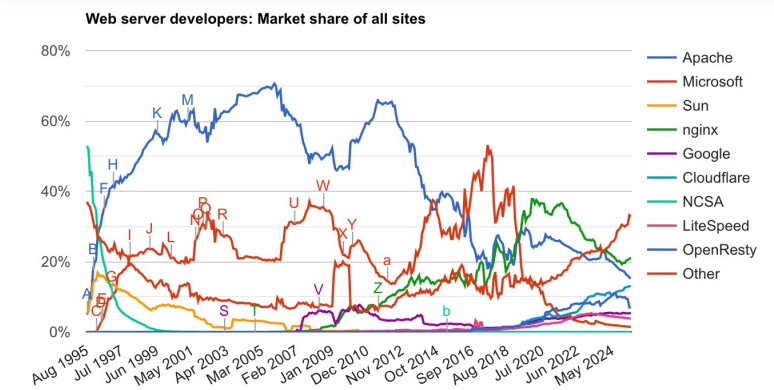
- Volume: 33 EB (22 fixed + 11 mobile) per day
- MAMAAN (Meta, Alphabet, Microsoft, Amazon, Alphabet and Netflix) generated 55% of Internet volume
- Video contributes 38% of downstream traffic
- Social media is the biggest app category in mobile networks' downstream and upstream traffic
- Expanding number of app categories and larger number of apps, which are producing more data

Source: https://www.sandvine.com/phenomena

Web growth



2014: 1G websites!



Metcalfe's law

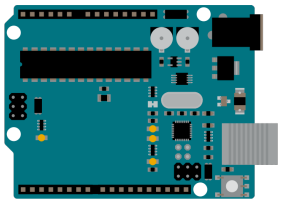
"The value of a telecommunications network is proportional to the square of the number of connected users of the system".

➡ Networking is *socially* and *economically* interesting

Computing power

- Computers got…
  - Faster
  - Cheaper
  - Power efficient
  - Smaller

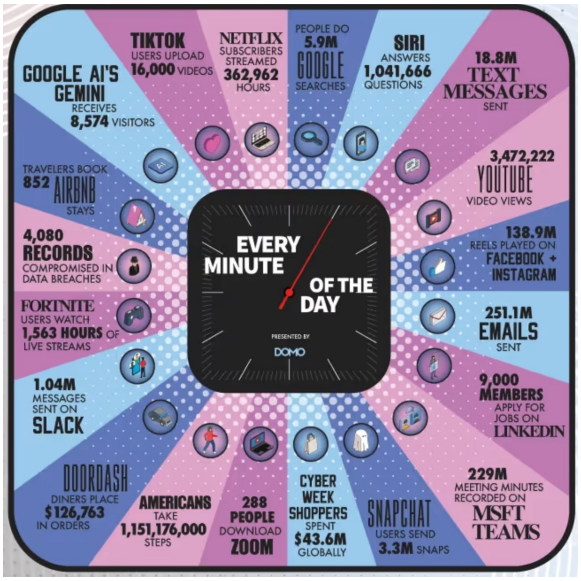
- 1974: Intel 8080
  - 2 MHz, 6K transistors
- 2004: Intel P4 Prescott
  - 3.6 GHz, 125 million transistors
- 2011: Intel 10-core Xeon Westmere-EX (multicore CPUs)
  - 3.33 GHz, 2.6 billion transistors
- 2019: NVIDIA Turing GPU
  - 14.2 TFLOPS of peak single precision (FP32) performance



Arduino UNO: weight=25 g, width=53.4 mm, length=68.6 mm

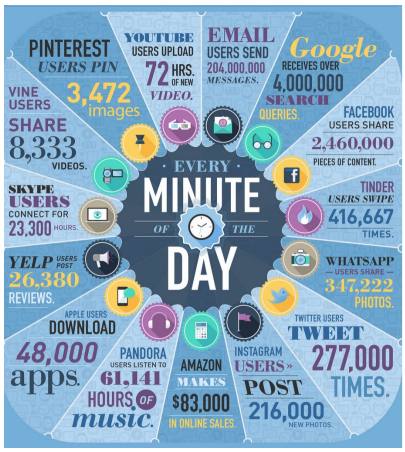
Every minute in Internet

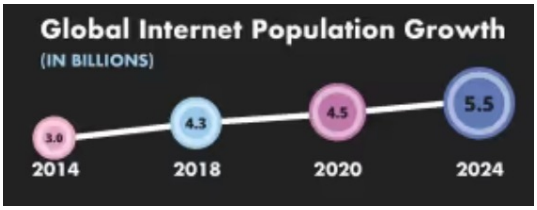
- Huge data volume (2024)



https://www.domo.com/learn/infographic/data-never-sleeps-12

- Evolution (how it was in 2014)





The challenge: scale of Internet services

- How to provide and manage these services?
- It is a problem of scale and scale changes everything!



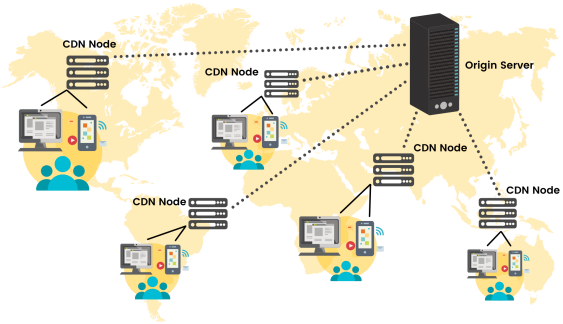
Google in 1997

Scale *has been the single most important force driving changes in system software over the last decade, and this trend will probably continue for the next decade.*

Ousterhout, Is scale your enemy, or is scale your friend?: technical perspective, *Comm. ACM*, 2011
https://dl.acm.org/doi/pdf/10.1145/1965724.1965748

Distributed systems: not only Internet and Web

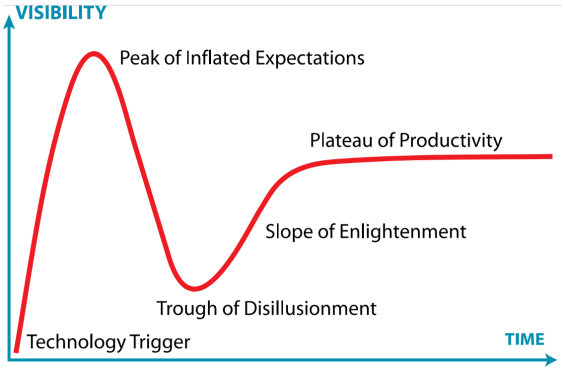
- Internet and Web: notable examples of distributed systems



- Other examples:
  - Cloud systems, HPC systems
  - Peer-to-peer systems
  - Internet of Things (IoT)
  - Smart home networks



Gartner's IT hype cycle for emerging technologies

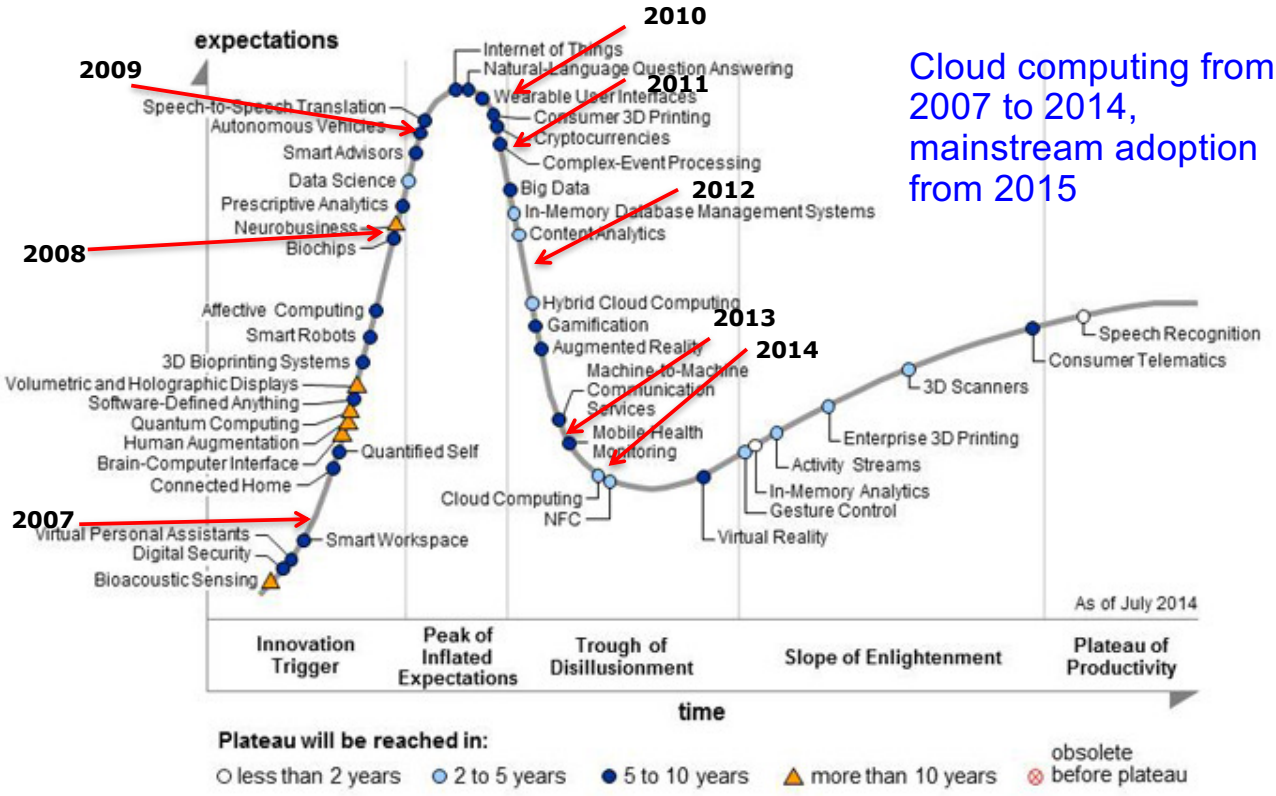


Graphical presentation developed by IT firm Gartner to represent maturity, adoption, and application of specific technologies

5 phases:

1) Innovation trigger: a new technology is introduced
2) Peak of inflated expectations: hype tends to be overhyped, with high and often unrealistic expectations about technology impact
3) Trough of disillusionment: interest wanes as reality sets in, possible decline in investment
4) Slope of enlightenment: more practical applications start to be found, with a more balanced and clearer understanding of its potential
5) Plateau of productivity: technology widely adopted, reaches its full potential, becoming a mainstream tool

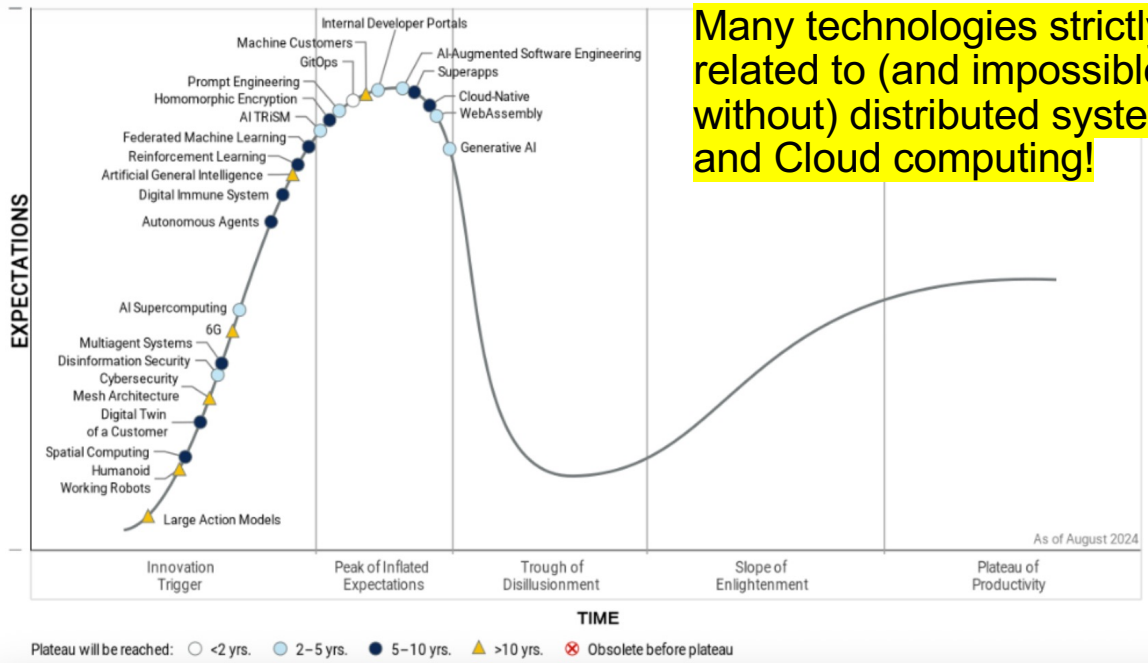
(Old) Gartner's hype cycle for emerging technologies and cloud



Cloud computing from 2007 to 2014, mainstream adoption from 2015

2024 Gartner's hype cycle for emerging technologies

Figure 1. Hype Cycle for Emerging Technologies, 2024



Many technologies strictly related to (and impossible without) distributed systems and Cloud computing!

Distributed systems, Cloud and AI

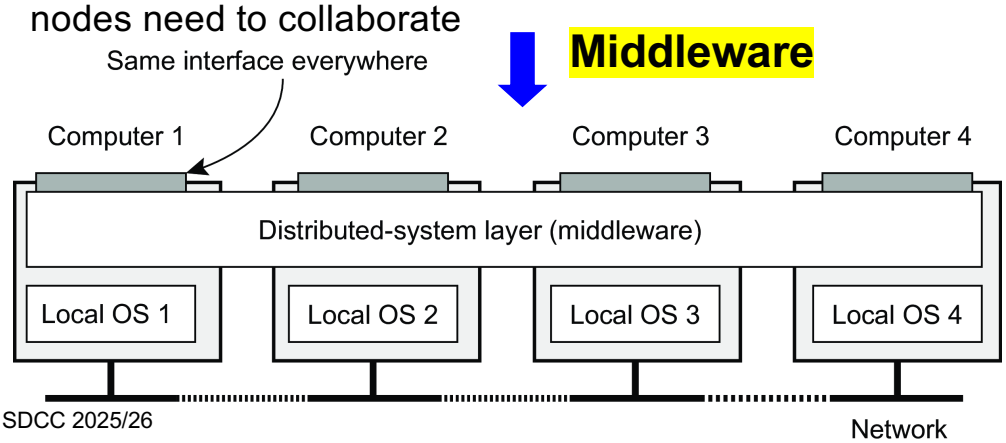
- AI has recently become practical as result of:
  - Distributed computing
  - Affordable cloud computing and storage costs
  - Examples: federated learning, distributed training of foundation models (huge computational and data storage needs)
- Distribute = to divide and dispense in portions
- A foremost strategy used in distributed computing you already know
  - Divide et impera: break larger (computational) problems down into numbers of smaller, interrelated, “manageable” pieces

Distributed system

- Multiple definitions of **distributed system (DS)**

[van Steen & Tanenbaum] A distributed system is a collection of autonomous computing elements that appears to its users as a single coherent system

  - Consists of autonomous computing elements (i.e., **nodes**), can be hardware devices (computer, phone, car, robot, …) or software processes
  - Users or applications perceive it as a **single system** (how?): nodes need to collaborate



Distributed system

[Lamport] A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable

- Emphasis on fault tolerance

- Who is Leslie Lamport?
  - Recipient of 2013 Turing award https://www.youtube.com/watch?v=pgWTmOyUjtM
  - His research contributions have laid the foundations of theory and practice of DS
    - Fundamental concepts such as causality, logical clocks and Byzantine failures
    - Algorithms to solve many fundamental problems in DS
  - Inventor of LaTeX

Why make a system distributed?

- Share resources
  - Resource = computing node, storage, data, service, …
- Lower costs
- Improve performance
  - e.g., get data from a nearby node rather than one halfway round the world
- Improve availability and reliability
  - even if one node fails, the system as a whole keeps functioning
- Improve security
- Solve bigger problems
  - e.g., huge amounts of data, can't fit on one machine
- Support Quality of Service (QoS) and Quality of Experience (QoE)

Why to study distributed systems

- Distributed systems are more complex than centralized ones
  - e.g., no global clock, group membership, …
- Building them is harder… and building them correct is even much harder

  Distributed systems need radically different software than centralized systems do (Tanenbaum)
- Managing, and, above all, testing them is difficult

Some distinguishing features of DS

- Concurrency
  - Many things happen “at the same time”
  - Centralized system: design choice
  - Distributed system: fact of life to be dealt with
- No global clock
  - Centralized system: use computer’s physical clock for synchronization
  - Distributed system: many physical clocks and not necessarily synchronized among them
- Independent and partial failures
  - Centralized system: fails completely
  - Distributed system: fails partially (i.e., only a part), often due to communication; hard (and in general impossible) to hide partial failures and their recovery

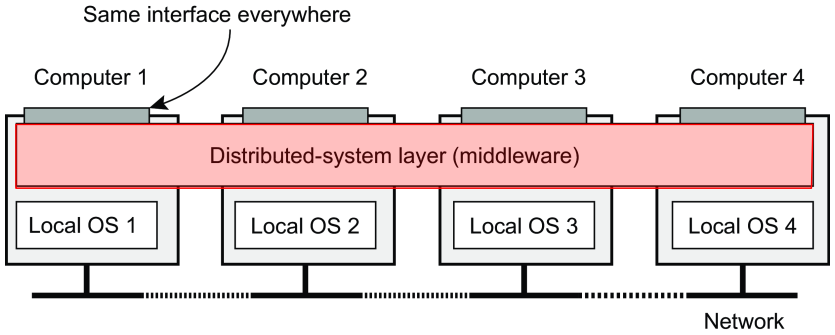
Challenges and design goals

- Challenges and goals associated with designing distributed systems
  1. **Heterogeneity**
  2. **Distribution transparency**
  3. **Openness**
  4. **Scalability**
  5. **Dependability**
  6. **Security**

while improving performance and sustainability (energy efficiency, carbon-awareness), reducing monetary cost, etc.

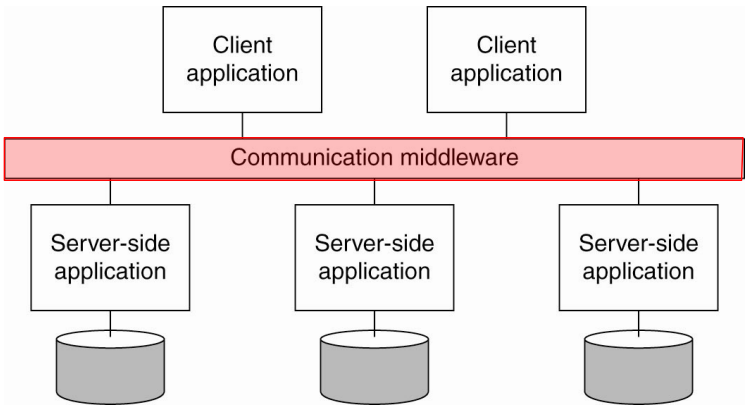
Challenge 1: Heterogeneity

- Many sources of heterogeneity: network, hardware, operating system (OS), programming language, implementations by different developers
- How to address? **Middleware**: the "OS of a DS"
  - Sw layer placed on top of OS that provides a programming abstraction as well as masks heterogeneity
  - Contains commonly used components and functionalities (e.g., communication) thus avoiding developers to implement them again and from scratch



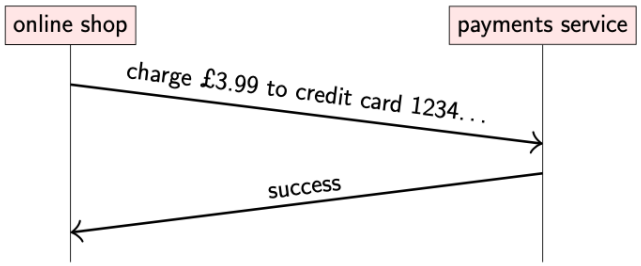
Communication middleware

- Facilitates communication among (heterogeneous) DS components/apps
- We will study
  - Remote Procedure Call (RPC)
  - Message Oriented Middleware (MOM)



Remote Procedure Call (RPC) example

- Online payment



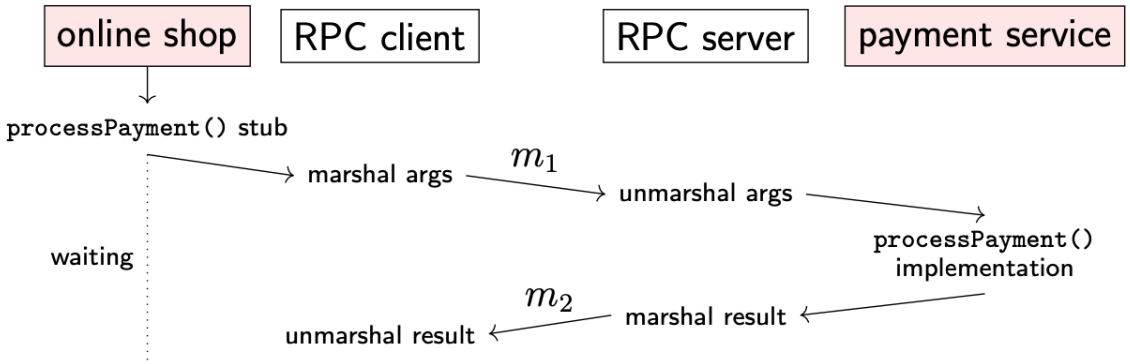

```
// Online shop handling customer's card details
Card card = new Card();
card.setCardNumber("1234 5678 8765 4321");
card.setExpiryDate("10/2024");
card.setCVC("123");

Result result = paymentsService.processPayment(card,
    3.99, Currency.GBP);

if (result.isSuccess()) {
    fulfilOrder();
}
```

Implementation of this function is on another node!

RPC: Behind the curtains



$m_1 =$

```
{
  "request": "processPayment",
  "card": {
    "number": "1234567887654321",
    "expiryDate": "10/2024",
    "CVC": "123"
  },
  "amount": 3.99,
  "currency": "GBP"
}
```

$m_2 =$

```
{
  "result": "success",
  "id": "XP61hHw2Rvo"
}
```


Challenge 2: Distribution transparency

- **Distribution transparency**: single coherent system where distribution of its *objects* (*resources* and *processes*) is *transparent* (i.e., invisible) to users and applications
- **Types** of distribution transparency (*ISO 10746*, Reference Model of Open Distributed Processing)

Access transparency

- Hide differences in data representation and how an object is accessed
  - e.g., use same mechanism for local or remote call

Location transparency

- Hide where an object is located
  - e.g., URL hides IP address

Relocation transparency

- Hide that an object may be moved to another location *while in use*

Challenge 2: Distribution transparency

Migration transparency

- Hide that an object may move to another location
  - e.g., communication between mobile phones

Replication transparency

- Hide that an object may be replicated
  - How? Same name for all replicas
  - Example: type command `dig www.youtube.com`
  - Require also location transparency

Concurrency transparency

- Hide that an object may be shared by several users
  - E.g.: concurrent accesses to DB table
  - Issue: leave shared object in a consistent state, e.g., by *locking* mechanisms

Failure transparency

- Hide failure and recovery of an object (Lamport's DS definition)

Degree of distribution transparency

- Aiming at *full* distribution transparency may be too much
  - We cannot always hide communication latency: sending a message from Roma to New York requires ~23 ms
  - We cannot completely hide failures in a large-scale DS
    - Cannot distinguish a slow server from a failing one
    - Cannot be sure that a server actually performed an operation before crashing
  - Price for achieving full transparency may be too high in term of performance
    - e.g., keeping data replicas *exactly* up-to-date takes time
    - e.g., *immediately flushing* write operations to disk for fault tolerance affects performance

  Trade-off between consistency and performance

Challenge 3: Openness

- Open DS: offers components that can easily be used by or integrated into other systems; consists of components that originate from elsewhere
- System should conform to well-defined interfaces
  - Defined through IDL (**Interface Definition Language**)
    - IDL is used to define data structures, interfaces (APIs), serialization formats in a language-agnostic way: complete and neutral
    - IDL examples: Protobuf, XDR, Thrift
- Benefits of openness
  - Interoperability
  - Portability of applications
  - Extensibility
- Examples: Jakarta EE, .Net, Web Services

Practice shows that many distributed systems are not as open as we'd like (van Steen & Tanenbaum)

Separation of policies and mechanisms

- To implement open and flexible DS, we need to organize the DS as a collection of relatively small and easily replaceable or adaptable component rather than as a monolithic system
- How? Separate policies from mechanisms
  - Policies refer to the principles guiding the decisions, while mechanisms represent the means to implement policies
- E.g., caching in a web browser:
  - Mechanism: store data and allow (dynamic) setting of caching policies
  - Caching policies:
    - Where to cache data?
    - How to free space when cache fills up?
    - When to refresh cached data?
    - Private or shared cache?

Separation of policies and mechanisms

- The other side of the coin
  - Strict separation can be counterproductive: the stricter the separation between policies and mechanisms, the more we need to ensure proper mechanisms, potentially leading to many configuration parameters and complex management
- Need to find a balance
- Possible solution: *self-configurable systems*

> *Finding the right balance in separating policies from mechanisms is one of the reasons why designing a distributed system is sometimes more an art than a science* (van Steen & Tanenbaum)

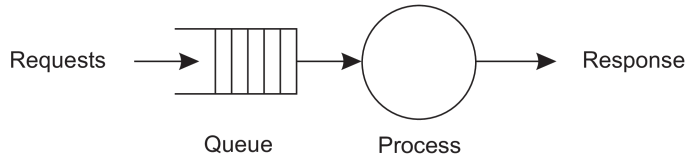
Challenge 4: Scalability

- **Scalability** is the property of a (distributed) *system* to keep an adequate level of performance notwithstanding a growing amount of:
  - Number of users and resources (**size** scalability)
  - Maximum distance between nodes (**geographical** scalability)
  - Number of administrative domains (**administrative** scalability)
- Most systems account only, to a certain extent, for size scalability

> *Many developers of modern distributed systems easily use the adjective scalable without making clear why their system actually scales.* (van Steen)

Size scalability

- Why a centralized system often lacks scalability
  - Computational capacity, limited by CPUs
  - Storage capacity, including transfer rate between CPUs and disks
  - Network between users and centralized system
- Formal analysis (see PMCS course)

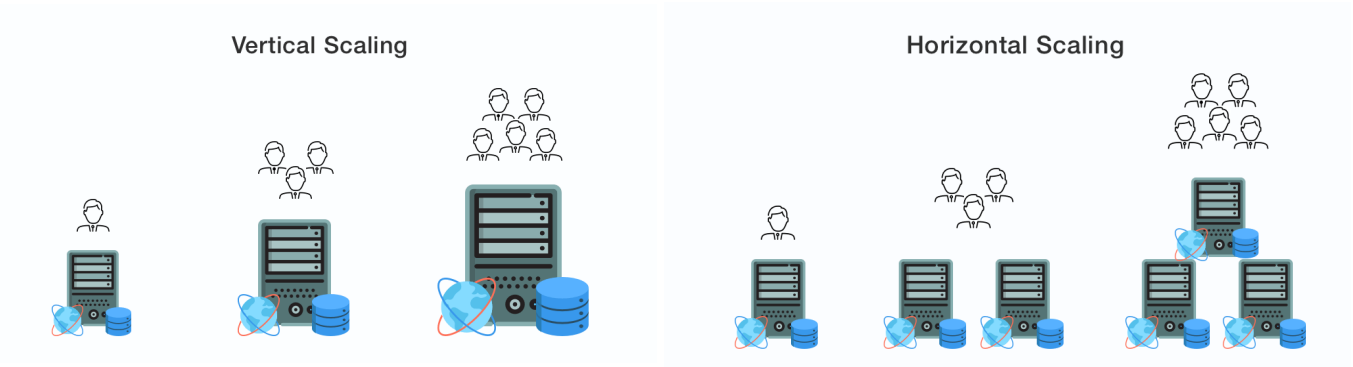


$$\frac{R}{S} = \frac{1}{1-U}$$

  - Service time S
  - Utilization U: fraction of time the service is busy
  - Response time R: total time take to process a request its arrival
  - If U is small, response-to-service time is close to 1: request is immediately processed
  - If U goes up to 1, system comes to a grinding halt. Solution: decrease S. How?

Size scalability

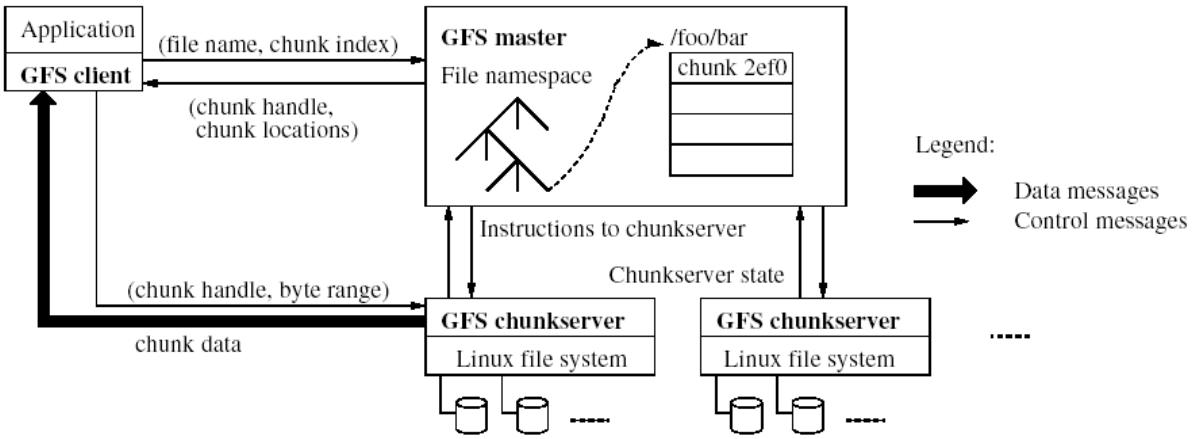
- Two directions for size scalability
  - **Vertical (scale-up)**: more powerful resources
  - **Horizontal (scale-out)**: more resources with same capacity



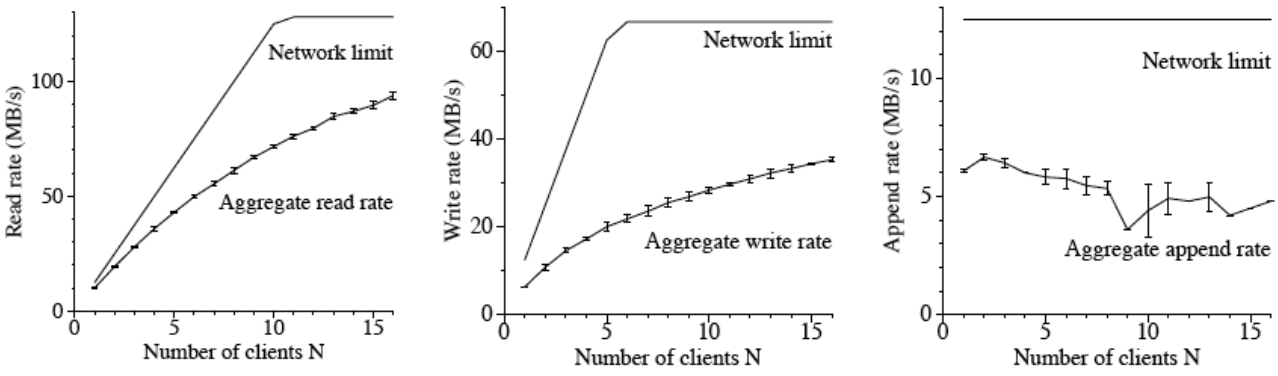
- Which resources? Physical, virtual (virtual machines, software containers, etc.)

Size scalability: GFS example

- Google File System (GFS)
  - Distributed file system based on divide et impera principle and horizontal scalability
  - Read, write, append operations
  - File divided into fixed blocks (chunks), which are distributed and replicated on multiple servers
  - Single master, multiple workers (chunkservers)



Size scalability: GFS example



(a) Reads (b) Writes (c) Record appends

- See experimental results
  - Scale parameter: number of clients
  - Scalability metric: aggregated read/write/append throughput (random file access)
  - Scalability criterion: the closer to network limit, the better
- Scalability is challenging!

Scaling techniques

1. Hide communication latency
   - Use asynchronous communication
   - Have separate handler for incoming response
   - Problem: not every application fits this model

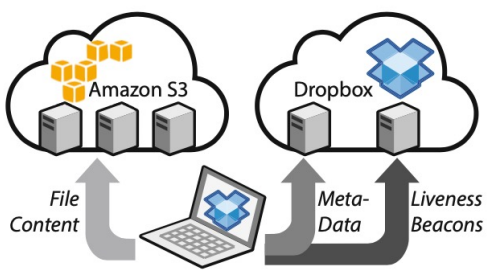
2. Move computation to clients

3. Distribute data and computation across multiple resources
   - Divide et impera: partition data and computation into smaller parts and distribute them across multiple DS resources
   - E.g., decentralized naming service (DNS), data-intensive distributed computation (Hadoop MapReduce and Spark)

Scaling techniques

4. Replicate DS resources and data
   - Make copies of data available at different DS nodes
   - Distribute processing on multiple DS nodes
   - Examples:
     - Distributed file systems and databases
     - Replicated Web servers
     - Web caches (in browsers and proxies)

- Example: in a cloud storage service (e.g., Dropbox, OneDrive, GDrive) data are locally cached on your device and replicated across multiple cloud servers



The problem with replication

- Applying replication is easy, but
- Having multiple copies leads to inconsistency: modifying one copy makes that copy different from the rest
- Trade-off: depending on application type, a certain degree of inconsistency can be tolerated
  - Blog, shared file, shopping cart, on-line auction, air traffic control
- We will study different consistency models to choose from

Challenge 5: Dependability

- **Dependability** refers to the degree that a computer system can be relied upon to operate as expected
  - partial failures make it intricate for distributed systems
- Requirements related to dependability
  - **Availability**: readiness for usage
  - **Reliability**: continuity of service delivery
  - **Safety**: very low probability of catastrophic consequences
  - **Maintainability**: how easily a failed system can be repaired

Dependability: availability

Can I use the system now?

- System is ready to use (i.e., operational) immediately
- Availability $A(t)$ of component C: probability that C is functioning correctly at time t

- Availability = uptime / (uptime + downtime)

- Normally expressed as number of 9's https://uptime.is/
  - A = 99%: two nines

    1-A=1% average days per year

    downtime per year = 0.01*365.2425 d = 3d 15h 39m 29.5s
  - A = 99.99%: four nines

    downtime per year = 0.0001*365.2425 d = 52 m 35.7 s

Dependability: availability

- Metrics

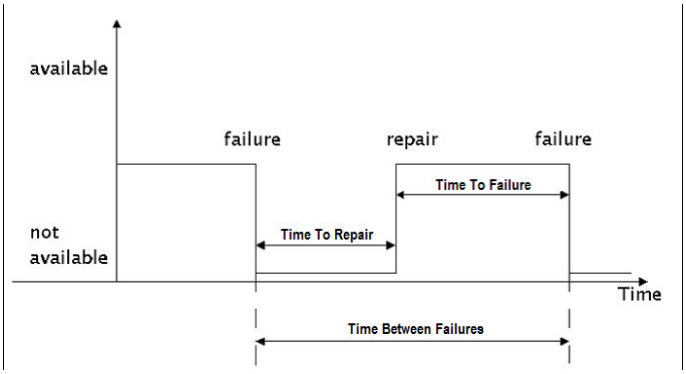
  A = MTTF/(MTTF + MTTR)

  Mean Time To Failure (MTTF): average time until a component fails

  Mean Time To Repair (MTTR): average time needed to repair a component

  Mean Time Between Failures (MTBF): MTTF + MTTR

  MTBF = total operating time / number of failures



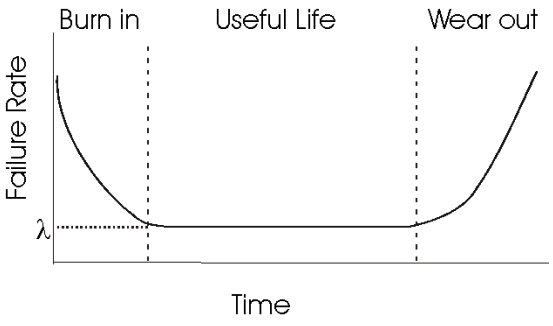
Dependability: reliability

Will the system be up as long as I need it?

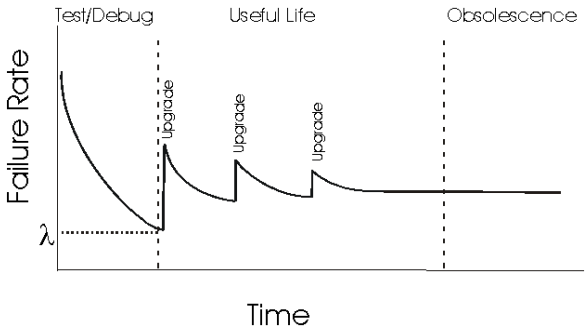
- System will run continuously without failure
- Reliability $R(t)$ of component C: conditional probability that C has been functioning correctly during $[0,t)$ given C was functioning correctly at the time $T = 0$

- Metrics: MTTF (and failure rate)

Bathtub curve for hardware



Revised bathtub curve for software



Availability vs reliability

- Availability ≠ reliability (when the system is repairable)

- Example A: system that goes down for 1 ms every hour
  - Highly available: > 99,9999% (= 1 - 1/(3600*1000))
  - Unreliable, because MTBF = 1 hour and there are 24*365=8780 failures per year

- Example B: system that never crashes but is shutdown for 2 weeks every year
  - Highly reliable, because MTBF = 1 year and there is only 1 failure per year
  - But only 96% available (= 1 - 14/365)

Failure, error and fault

- **Failure**: the system (or a component) is not living up to its specifications
  - Example: software crashes or produces wrong result
  - Can be cascading
- **Error**: wrong internal state, may lead to failure
  - Example: memory stores wrong value
- **Fault**: cause of an error
  - Example: memory is damaged
  - Can be transient, intermittent, permanent
- Relationship: fault $\rightarrow$ error $\rightarrow$ failure

Dependability: tools

- **Fault prevention**
  - Prevent fault occurrence
- **Fault tolerance**
  - Build component and make it mask fault occurrence
- **Fault removal**
  - Reduce presence, number, or seriousness of faults
- **Fault forecasting**
  - Estimate current presence, future incidence, and consequences of faults

Challenge 6: Security

- A distributed system that is not secure, is not dependable
- What we need
  - **Confidentiality**: information is disclosed only to authorized parties
  - **Integrity**: ensure that alterations to system assets can be made only in an authorized way
- Authorization, authentication, trust
  - Authentication: verifying correctness of claimed identity
  - Authorization: does an identified entity has proper access rights?
  - Trust: one entity can be assured that another will perform particular actions according to a specific expectation

Security mechanisms

- Keeping it simple: it is all about encrypting and decrypting data using security keys
- Symmetric vs asymmetric cryptosystem
  - Symmetric: same encryption and decryption key, both keys should be kept secret
  - Asymmetric: public key and private key
- Secure hashing
  - In practice, we use secure hash functions: $H(data)$ returns a fixed-length string
    - Any change from *data* to *data*$*$ will lead to a completely different string $H(data*)$
    - Given a hash value h, it is computationally infeasible to find a *data* with $h = H(data)$

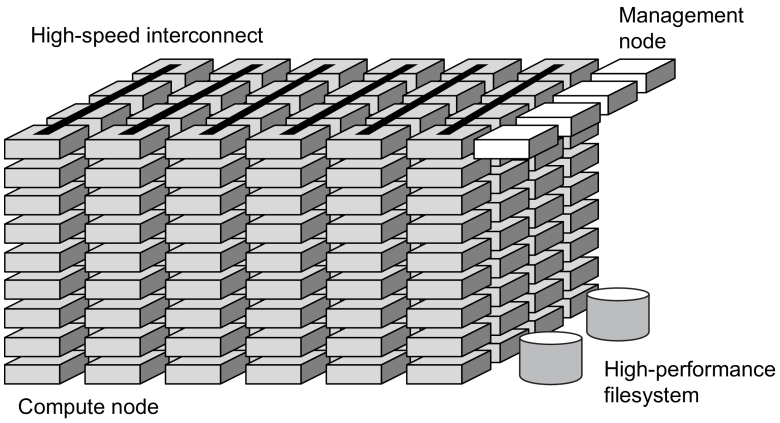
Some distributed systems

- High-performance distributed computing systems
  - Cluster computing
  - Cloud computing
  - Edge computing

- Distributed pervasive systems

Cluster computing

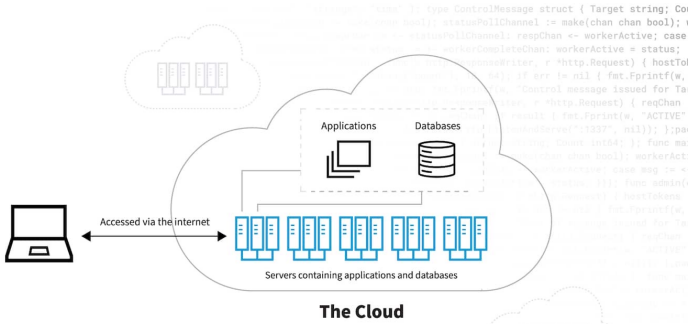
- Cluster: group of high-end systems connected through a LAN
  - Typically homogeneous: same OS, near-identical hardware
  - Single (or some tightly coupled) managing node(s)



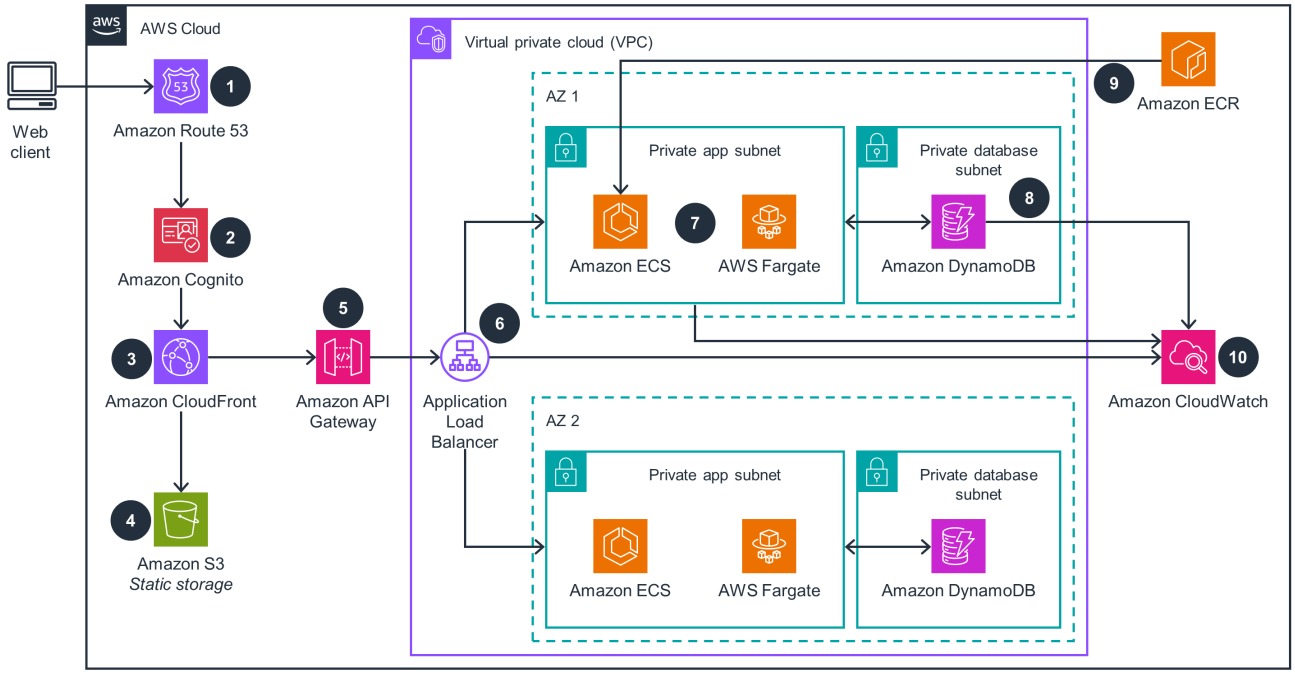
- Clusters dominate TOP500 architectures
  https://www.top500.org/

Cloud computing

- Infrastructures, platforms and software are managed by cloud providers
- Users only lease cloud services on demand and are billed on a pay-as-you-go model
- With respect to cluster, cloud is:
  - available to anyone
  - on a much wider scale
  - does not require users to physically own or use hardware



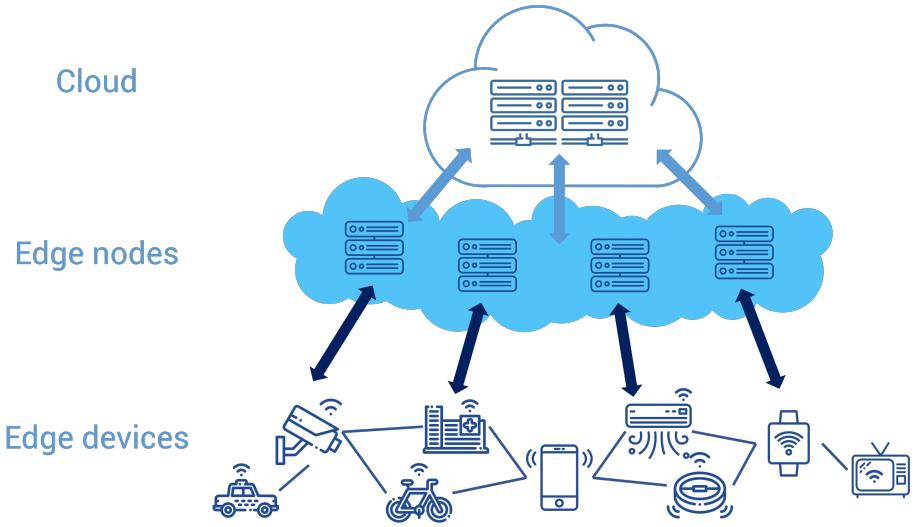
Distributed application in cloud



https://aws.amazon.com/it/solutions/guidance/building-a-containerized-and-scalable-web-application-on-aws

Edge computing

- Brings computation and storage at the network edges, in proximity of data producers (e.g., IoT devices) and consumers (e.g., users)



Distributed pervasive systems

- Distributed systems where computing is embedded across many devices, which are
  - often small, mobile, embedded, battery-powered
  - often spread out geographically
  - coordinated to work together seamlessly
    - The system naturally blends into the user's environment
- Sub-categories:
  - Ubiquitous computing systems: pervasive and continuously present (continuous interaction with users), e.g., smart home system
  - Mobile computing systems: pervasive with mobile devices, e.g., fleet of drones for emergency management
  - Sensor networks: pervasive, that work together to sense and monitor data about environmental or physical conditions, e.g., air quality

Pitfalls in realizing distributed systems

- Many distributed systems are needlessly complex because of errors in design and implementation that were patched later
- Common wrong assumptions by architects and designers of distributed systems ("The Eight Fallacies of Distributed Computing", Peter Deutsch, 1991-92):
  1. The network is reliable
     - "You have to design distributed systems with the expectation of failure" (Ken Arnold)
  2. Latency is zero
     - Latency is more problematic than bandwidth
     - "At roughly 300,000 km/s, it will always take at least 30 ms to send a ping from Europe to the US and back, even if the processing would be done in real time." (Ingo Rammer)
  3. Bandwidth is infinite

Pitfalls in realizing distributed systems

4. The network is secure
5. The topology does not change
   - That's right, it doesn't - as long as it stays in the test lab!
6. There is one administrator
7. Transport cost is zero
   - Going from the application level to the transport level is not free
   - Costs for setting and running the network are not free
8. The network is homogeneous

Technology is not the solution to everything!

References

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